CAPSTONE PROJECT

Connecta

Online Event Management System for a 2D Virtual World

B.Sc.IT ID66 | INT371 Experiential Learning Project I

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Team Member



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FRONTEND

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01

Overview


AURORA
Ideas for a brighter you
AURORA
Ideas
People
Impact
PIXEL PATH
Small Steps
Big Adventuros
PIXEL PATH
Play
Learn
Connect
GreenLoop
A Cleaner
Brighter Tomorrow
GreenLoop
People
Planet
Progress
Good
People
Brighter
Tomorrow
Together
Create
What's Next
Good
People
Brighter
Tomorrow
1:1 Meeting Room
MEET
SHARE
BELONG
Good
Ideas
Brighter
People
Same
People
New
Possibilities
Welcome !
Events
People
Ideas
Community

Project Track and Subtrack

TRACK 1

Application Development

SDGs

- Goal 10 Reduced Inequalities
  - People living upcountry cannot attend the event.
- Goal 13 Climate Action
  - Sustainable Development Goals
  - Driving Climate Action through an online event platform.

Reduced travel · Fewer on-site resources · Carbon impact measurement · Reduced Inequalities

Background & Problem Statement

1. Limitations of Online Events

Online platforms reduce location, cost, and distance constraints, but limit interaction and networking between participants.

2. The Missing Event Experience

Participants cannot naturally move, meet others, talk in small groups, or approach Speakers privately like in On-site events.

3. Gaps in Current Solutions

Virtual platforms mainly focus on Virtual Offices, while event management is often fragmented or requires manual coordination.

4. Connecta

Connecta is a 2D virtual event platform where participants can move, interact, join Sessions, and connect with Speakers and Booths, including a 1:1 Queue System.

5. Value & Target

Organizers can create and manage events independently and track participant engagement and carbon impact reduction.
Target: Regular event organizers with 30–80 participants.

Project Objectives

01 2D Virtual Event Space

Develop a 2D virtual event space with real-time movement and proximity communication for 30–80 users.

02 Interaction & Networking

Enable On-site-like interaction through 1:1 Speaker/Booth queues and group activities.

03 Self-service Event Management

Enable organizers to create and manage events, Sessions, Booths, and participants independently.

04 Post-event Analytics

Collect participation and interaction data to generate post-event reports and feedback.

05 Technical Feasibility

Evaluate Proximity Audio, concurrent users, bandwidth, and infrastructure costs.

06 Real-world Evaluation

Evaluate the system through at least one real event and gather user feedback.

How We Arrived At This Topic : Design Thinking

Empathize

Our own failed attempt to join a tech meetup online, then a survey to check it was not just us

Define

The one problem that needs a virtual world: small-group talk, private speaker access and in-event activity

IDEATE

Three approaches compared; 2D world plus event management solves both problem levels

PROTOTYPE

Technical spikes run before proposing the topic

TEST

Low-fidelity flows checked with people who have never used a system like this

Differentiators

Virtual event experience

Walk the venue, see who is where, and join the conversation you choose

Self-serve

Create an event and know the price in ten minutes, without contacting sales

Event life cycle

Sessions, tickets and booths are objects in the system, not announcements made elsewhere

Analytics & Report

Turnout, dwell time and interest per session and booth, ready for the next event

Networking

Contacts exchanged as virtual namecards, exported to your phone after the event

Queue Management

Ordered one-on-one meetings with speakers and booths, with no burden on the speaker

Development Challenges

01 AUDIO

Peer-to-peer voice breaks as avatars move; an SFU changes who you hear without reconnecting

02 NETWORK

The defense runs on the most restricted network we will meet, so TURN over TLS is mandatory

03 BANDWIDTH

Fifty cameras is impossible, so everyone is an avatar and video stays on stage and one-on-one

04 PERMISSIONS

Five roles across many events on one system, enforced server-side and tested for cross-event access

05 ONBOARDING

Attendees who cannot work out the controls in minutes leave, and the technical work stops mattering

06 CAPACITY

Three people, none experienced in realtime or voice, so estimates are tripled and scope is ordered

We Already Spike

SPIKE 1

Voice over SFU

LiveKit 8–10 users quality and stability

SPIKE 2

realtime sync

Position synchronization and latency for approximately 30 users

SPIKE 3

2D rendering

Tiled + Phaser + collision + zoning with Nextjs

Technical Scope

“About seventy percent is web work the team has built before”

Preliminary Feasibility Study

MARKET

Virtual event platforms still growing at 17.4% a year through 2030

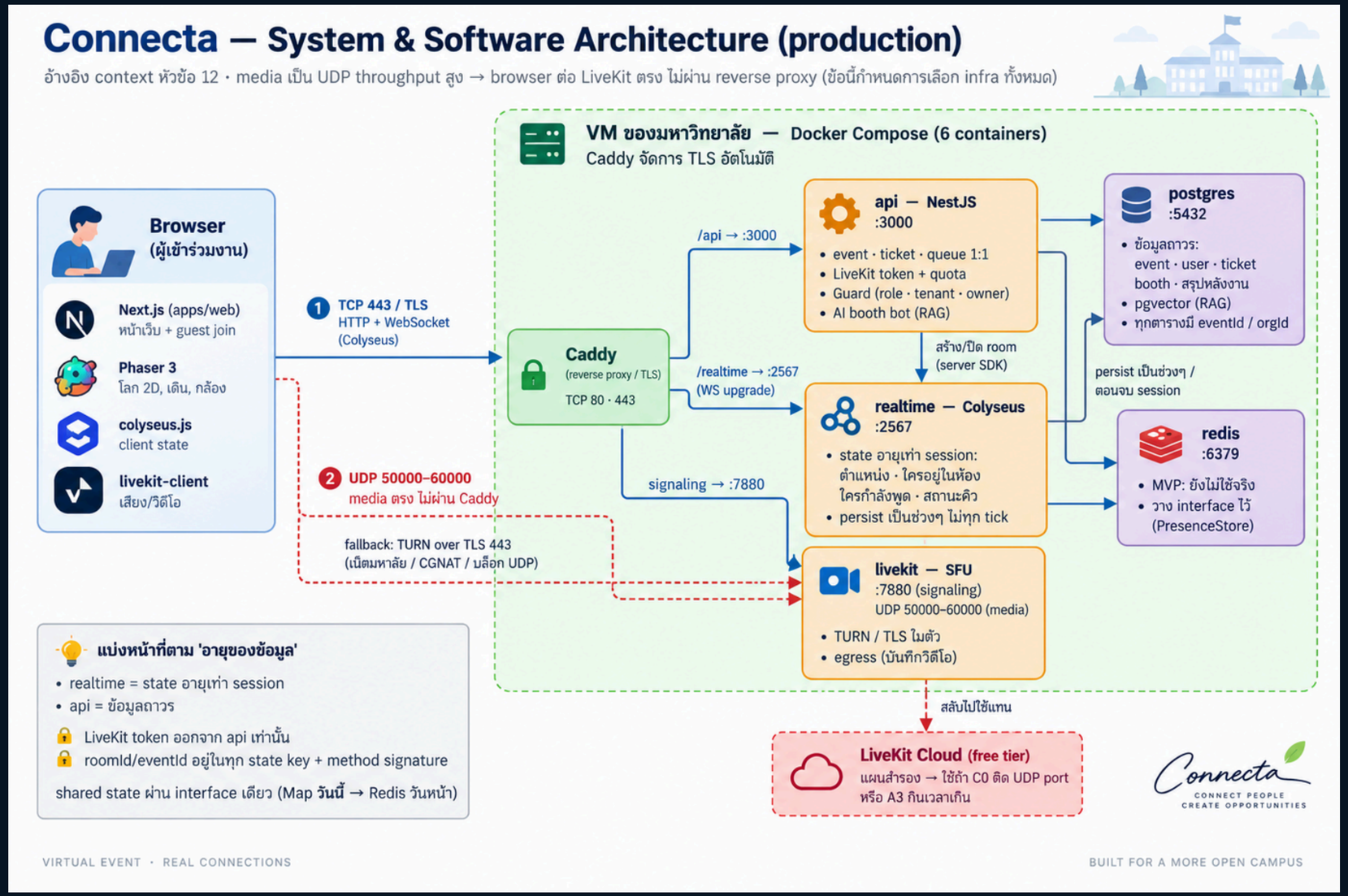
DEMAND

Networking is the top reason people attend, and the thing online events do worst

EVIDENCE

Foreign data establishes the problem; our own survey and interviews establish it here

System Overview and Architecture



System Features Overview · Features

DIGITAL BUSINESS CARD EXCHANGE

- Exchange digital business cards with other participants
- Save contact information for future communication

EVENT MAP MANAGEMENT

- Configure event maps and select a map for each event
- Set the event schedule and map availability status: Preparing, Open, or Closed
- Set participant capacity and view the corresponding event fee
- Configure event access: require registration before entry, allow public access, or require a ticket code

REGISTRATION & TICKET ACCESS

- Sign in with a Google or Microsoft account before joining an event
- Register by completing the required registration form
- Join public events or enter a ticket code for invitation-only events

System Features Overview · Zone

EVENT STAGE

Speaker presentation and moderated Q&A

- Speakers share video and speak
- Participants raise their hand to request to speak
- The active speaker grants temporary microphone access

BOOTH

Virtual exhibition space for exploring event content

- Participants discover information, products, or activities
- Booth staff introduce content and answer questions
- AI Assistanct help staff for answer viewer
- Anyone can open microphone in booth area



MEETING ROOM

Open discussion space, similar to Zoom

- Speakers and participants can turn on their microphones and talk freely
- Includes 1:1 Meeting Rooms for private conversations

Conversation Mode

Users can start a voice conversation within an area around their avatar. Other users can hear the conversation when they move into the zone.

- Public Mode: Users who enter the area can select Join to hear and participate in the conversation.
- Private Mode: The zone host must invite users before they are allowed to select Join and enter the conversation.



System Features Overview · Role

PLATFORM ADMIN

Oversee events · Monitor services

- View all events and suspend those that violate platform rules
- Review event activity through logs
- Monitor concurrent users and communication service status

ORGANIZER

Design events · Manage operations · Security

- Create events and define schedules and participant capacity
- Manage roles and permissions
- Event Status Management
- Mute / Kick / Ban

ATTENDEE

Enter the virtual world · Meet people · Build connections

- Join using an account
- Edit digital business cards
- Send public chat messages that appear above the sender’s avatar
- View participant profiles and request to exchange digital business cards

System Features Overview · Role

SPEAKER

Lead sessions and engage with participants

- Lead sessions on the Stage
- Approve temporary microphone access for participants in the Stage zone
- Open Meeting Rooms and set participant capacity

EXHIBITOR

Manage and support booth experiences

- Run booth activities and sessions
- Configure booth layout and content
- Provide information and scripts for the AI assistant to answer participant questions

STAFF

Support event operations and moderation

- Monitor event activities and assist participants
- Moderate interactions and manage reports
- Add and assign speakers or booth staff role
- Mute, kick, or ban users when needed

CAPSTONE PROJECT

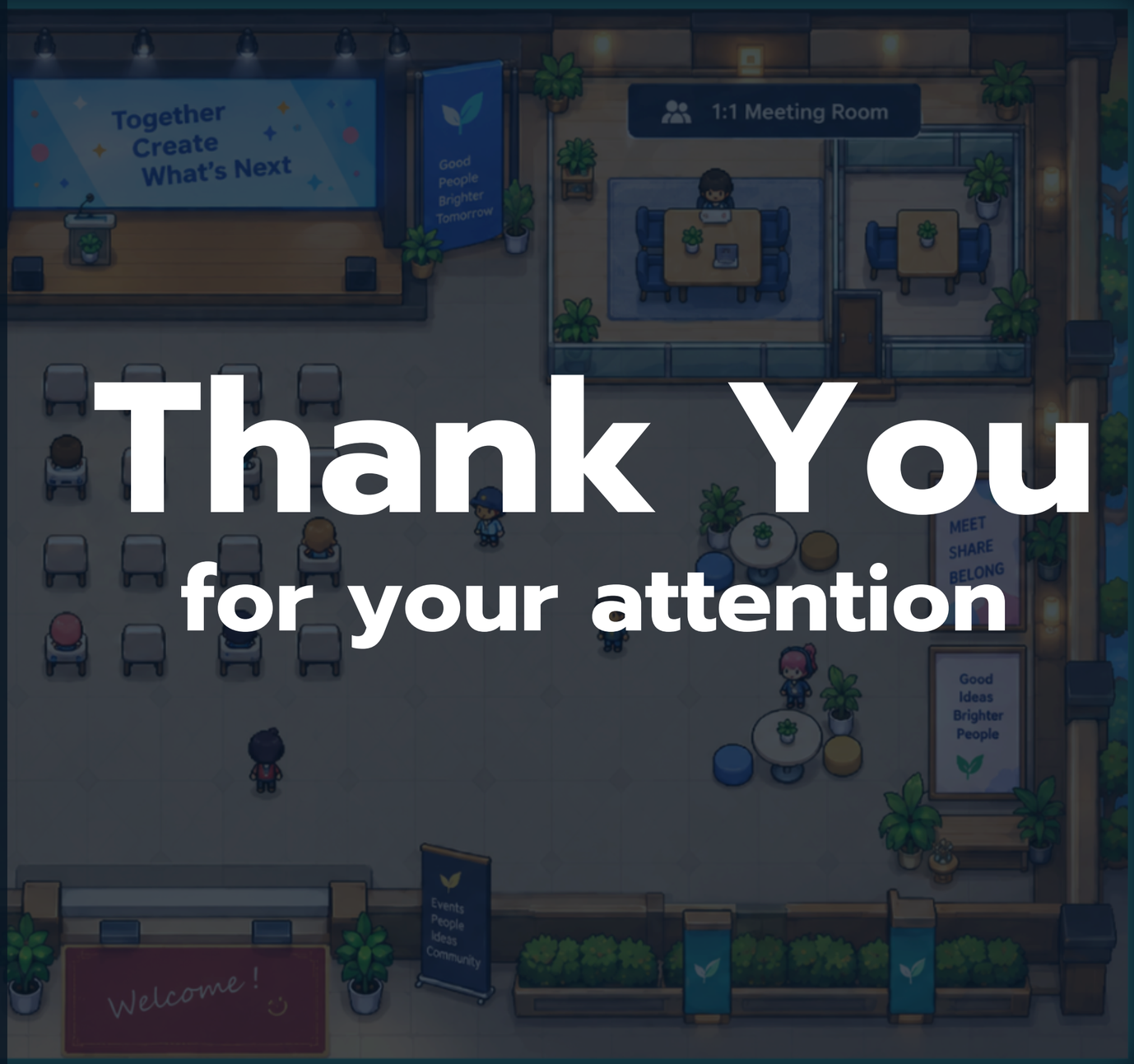
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Thank You

for your attention